

CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
222TCS100	ADVANCED DATA STRUCTURES AND ALGORITHMS	DISCIPLINE CORE 2	3	0	0	3

Preamble: The course introduces advanced data structures and algorithms in different domains. The goal of this course is to provide a solid background in the design and analysis of the major classes of algorithms. The course helps the learners to develop their own versions for a given computational task and to compare and contrast their performance.

Course Outcomes: After the completion of the course the student will be able to:*

CO 1	Analyse the relevance of amortized analysis and applications. (Cognitive Level: Apply)
CO 2	Illustrate string matching algorithms. (Cognitive Level: Apply)
CO 3	Illustrate advanced data structures like Binomial heap, Fibonacci heap, Disjoint set and string matching algorithms. (Cognitive Level: Apply)
CO 4	Illustrate network flow algorithms and applications. (Cognitive Level: Apply)
CO 5	Make use of probabilistic algorithms and approximation algorithms in computing. (Cognitive Level: Apply)
CO 6	Design, develop and implement software using advanced data structures and algorithms. (Cognitive Level: Create)

* The COs shown are only indicative. For each course, there can be 4 to 6 COs.

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	✓		✓	✓		✓	
CO 2	✓		✓	✓		✓	
CO 3	✓		✓	✓		✓	
CO 4	✓		✓	✓		✓	
CO 5	✓		✓	✓		✓	
CO 6	✓	✓	✓	✓	✓	✓	✓

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	80%
Analyse	20%
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design based questions (for both internal and end semester examinations).

Continuous Internal Evaluation : 40 marks

Micro project/Course based project : 20 marks

Course based task/Seminar/Quiz : 10 marks

Test paper, 1 no. : 10 marks

The project shall be done individually. Group projects not permitted.

Test paper shall include minimum 80% of the syllabus.

Course based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students.

End Semester Examination Pattern:

The end semester examination will be conducted by the University. There will be two parts; Part A and Part B. Part A contain 5 numerical questions with 1 question from each module, having 5 marks for each question. (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions.

Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

Total duration of the examination will be 150 minutes.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Explain how the accounting method of amortized analysis can be applied to stack operations.
2. Suppose we perform a sequence of n operations on a data structure in which the i^{th} operation costs i if i is an exact power of 2, and 1 otherwise. Use aggregate analysis to determine the amortized cost per operation.
3. What is the total cost of executing n of the stack operations PUSH, POP, and MULTIPOP, assuming that the stack begins with s_0 objects and finishes with s_n objects? Use potential method.

Course Outcome 2 (CO2)

1. Use an aggregate analysis to show that the running time of KMP-MATCHER is $\theta(n)$.
2. Working modulo $q = 11$, how many spurious hits does the Rabin-Karp matcher encounter in the text $T = 3141592653589793$ when looking for the pattern $P = 26$?
3. Compute the prefix function π for the pattern ababbabbabbababbabb

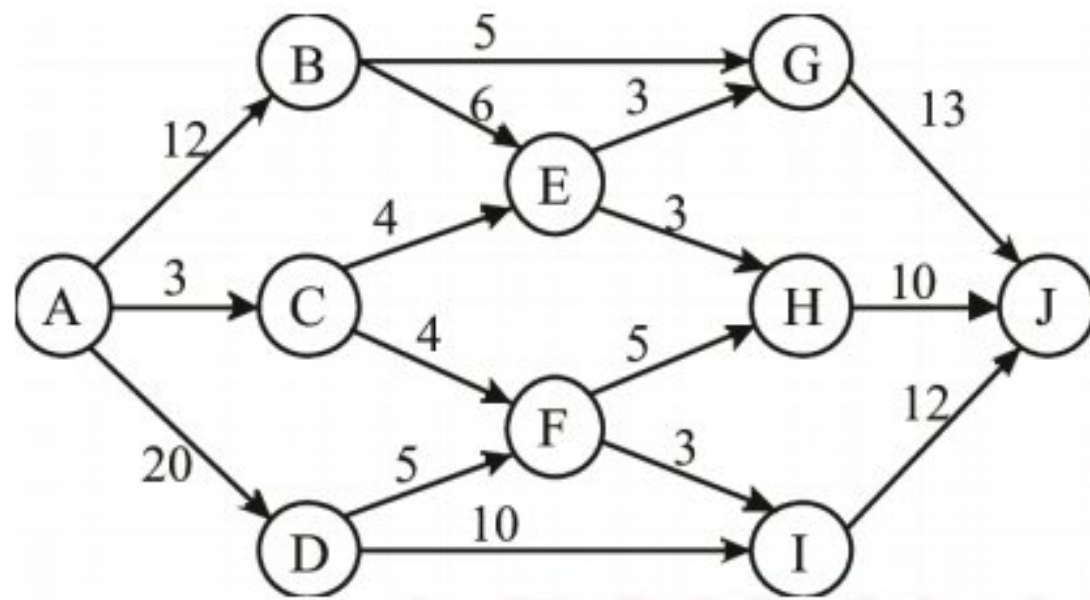
Course Outcome 3(CO3):

1. Analyse the time complexity of decrease-key operation of Fibonacci heap.
2. Illustrate extract-min operation of Binomial heap.

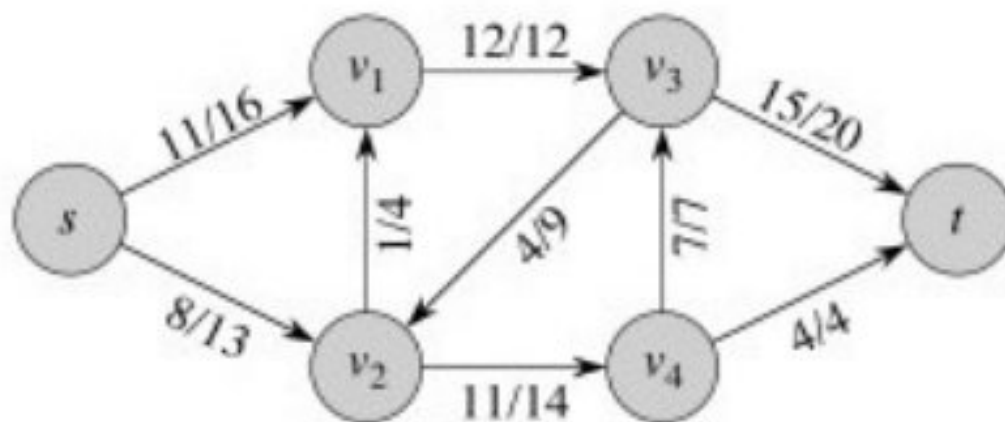
3. Explain the heuristics used in disjoint set data structure to improve the running time.

Course Outcome 4 (CO4):

1. Show the execution of the Edmonds-Karp algorithm on the given flow network (source: A and sink: J).



2. In the following figure, compute flow across the cut $(\{s, v_2, v_4\}, \{v_1, v_3, t\})$. What is the capacity of this cut?



3. State and prove max flow min cut theorem.

Course Outcome 5 (CO5):

1. Illustrate Miller-Rabin primality testing method.
2. Explain probabilistic selection algorithm.

Course Outcome 6 (CO6):

1. Explain the approximation algorithm for subset sum problem.
2. Consider each of the following words as a set of letters: $\{arid, dash, drain, heard, lost, nose, shun, slate, snare, thread\}$. Show which set cover GREEDY-SET-COVER produces when we break ties in favor of the word that appears first in the dictionary.

Model Question Paper

QP CODE:

Reg No: _____

Name: _____

PAGES : 4

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: 222TCS100

Course Name: ADVANCED DATA STRUCTURES AND ALGORITHMS

Max. Marks : 60

Duration: 2.5 Hours

PART A

Answer All Questions. Each Question Carries 5 Marks

1. Explain accounting method of amortized analysis with a suitable example.
2. Explain the algorithm for uniting two binomial heaps and analyse the running time.
3. Maximum matching in a bipartite graph G corresponds to a maximum flow in its corresponding flow network G' . Comment on this statement. Explain how maximum flow problem can be used to solve maximum bipartite matching problem.
4. Explain the probabilistic algorithm for verifying matrix multiplication problem.
5. Explain the approximation algorithm for traveling salesperson problem.

(5x5=25)

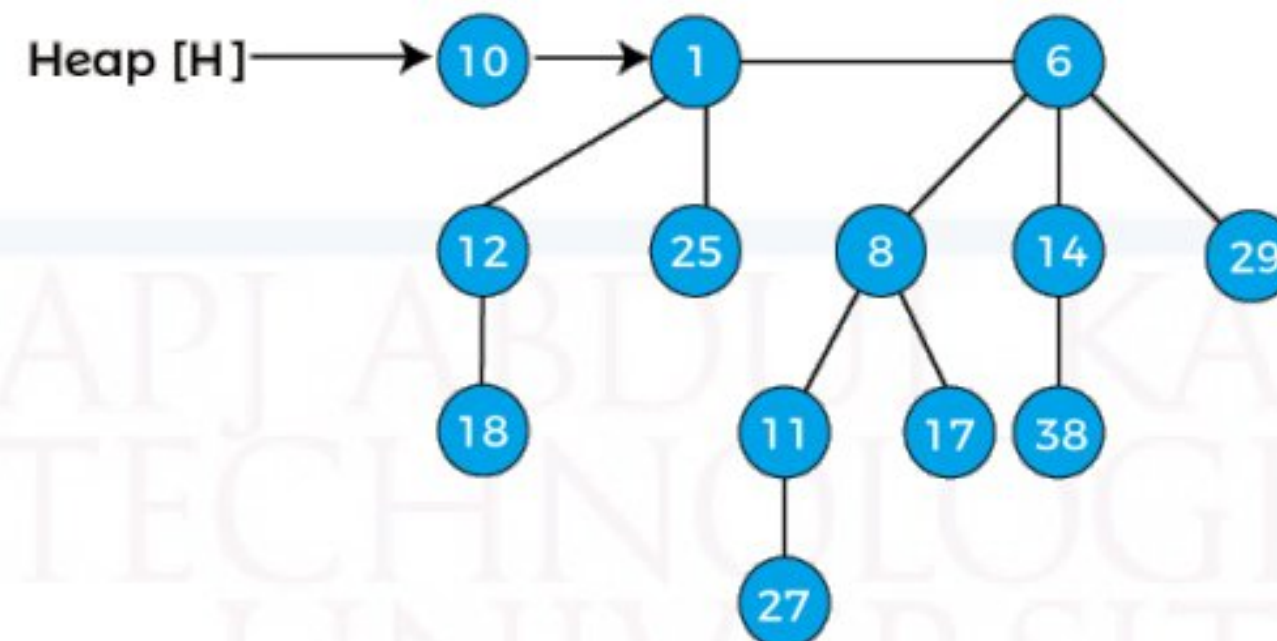
Part B

(Answer any five questions. Each question carries 7 marks)

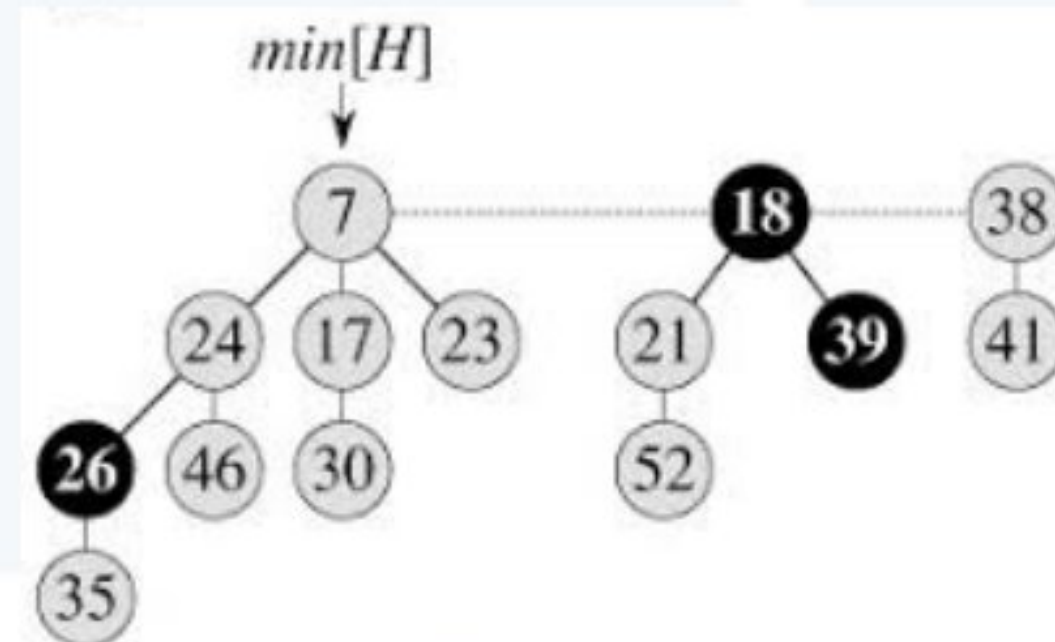
6. Describe Knuth-Morris-Pratt algorithm and illustrate using given text $T = \text{AABAACAADAABAABA}$ and pattern $P = \text{AABA}$. **(7)**
7. (a) Using potential method, compute the amortized cost of incrementing a binary counter. **(3)**
(b) Suppose we perform a sequence of n operations on a data structure in which the i^{th} operation costs i if i is an exact power of 2, and 1 otherwise. Use accounting method of amortized analysis to determine the amortized cost per operation. **(4)**
8. (a) Explain how disjoint set data structure is used to find connected components on an undirected graph. **(3)**

Model Question Paper

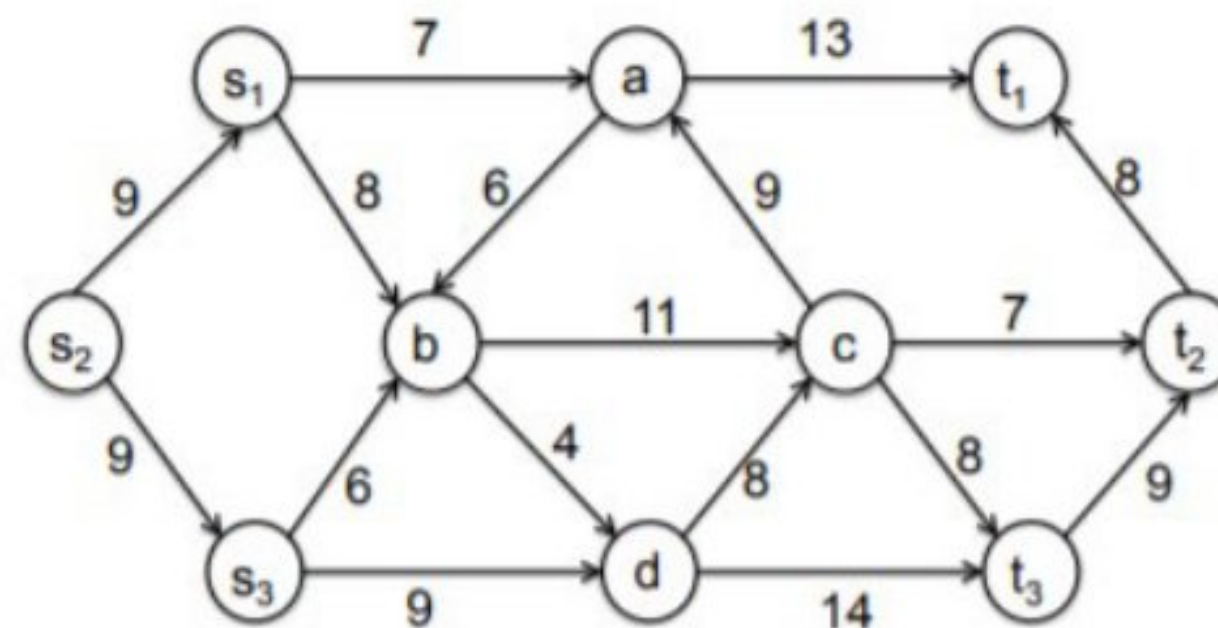
- (b) Show the binomial heap that results when a node with key 11 is deleted from the binomial heap shown in figure. (4)



9. (a) Explain the structure of Fibonacci heap. (2)
 (b) Apply extract minimum operation on the Fibonacci heap shown in figure and show the result. (5)



- 10 Describe Ford-Fulkerson algorithm and apply on the following network. Also obtain minimum cut across the network. (7)



- 11 (a) Apply Miller-Rabin algorithm to test whether the number 341 is prime or not. (4)
 (b) Explain probabilistic quick sort algorithm. (3)
- 12 (a) Describe polynomial-time approximation scheme and fully polynomial-time approximation scheme. (3)
 (b) Give an example of a graph for which APPROX-VERTEX-COVER always yields a suboptimal solution. (4)

Module – 1 (Amortized analysis and String matching)

Overview of asymptotic notations and complexity analysis, Amortized analysis – aggregate analysis, accounting method, potential method.

String matching – introduction, Rabin-Karp algorithm, Knuth-Morris-Pratt algorithm.

Module – 2 (Advanced data structures)

Overview of binary heap operations, Binomial tree and heap, Binomial heap operations, Fibonacci heap structure, Fibonacci heap operations, Disjoint set – overview, linked list representation, disjoint set forests.

Module – 3 (Network flow)

Network flow properties, examples, residual network, augmenting path, cut of network, maxflow-mincut theorem, Ford-Fulkerson algorithm, Edmonds-Karp algorithm, maximum bipartite matching.

Module – 4 (Probabilistic algorithms)

Introduction, types of probabilistic algorithms, Numerical algorithms – Numerical integration, Probabilistic counting, Monte-Carlo algorithms – Verifying matrix multiplication.

Number theory fundamentals – modular arithmetic, modular exponentiation, Euler's Theorem and Fermat's Theorem, Primality testing – Miller-Rabin test.

Las Vegas algorithms – Probabilistic selection and quick sort.

Module – 5 (Approximation algorithms)

Introduction, Vertex-cover problem, Traveling-salesman problem, Set-covering problem, Subset-sum problem.

Course Plan

No	Topic	No. of Lectures (37)
1	Module – 1 (Amortized analysis and String matching)	
1.1	Overview of asymptotic notations and complexity analysis	1
1.2	Amortized analysis – aggregate analysis	1
1.3	accounting method	1
1.4	potential method	1
1.5	String matching – introduction	1
1.6	Rabin-Karp algorithm	1
1.7	Knuth-Morris-Pratt algorithm (1)	1
1.8	Knuth-Morris-Pratt algorithm (2)	1
2	Module – 2 (Advanced data structures)	
2.1	Overview of binary heap operations	1
2.2	Binomial tree and heap	1

2.3	Binomial heap operations (1)	1
2.4	Binomial heap operations (2)	1
2.5	Fibonacci heap structure	1
2.6	Fibonacci heap operations (1)	1
2.7	Fibonacci heap operations (2)	1
2.8	Disjoint set – overview, linked list representation	1
2.9	disjoint set forests	1
3	Module – 3 (Network flow)	
3.1	Network flow properties, examples	1
3.2	residual network, augmenting path, cut of network	1
3.3	maxflow-mincut theorem	1
3.4	Ford-Fulkerson algorithm	1
3.5	Edmonds-Karp algorithm	1
3.6	Maximum bipartite matching	1
4	Module – 4 (Probabilistic algorithms)	
4.1	Introduction, types of probabilistic algorithms	1
4.2	Numerical algorithms – Numerical integration, Probabilistic counting	1
4.3	Monte-Carlo algorithms – Verifying matrix multiplication	1
4.4	Number theory fundamentals – modular arithmetic, modular exponentiation	1
4.5	Euler's Theorem and Fermat's Theorem	1
4.6	Primality testing – Miller-Rabin test (1)	1
4.7	Primality testing – Miller-Rabin test (2)	1
4.8	Las Vegas algorithms – Probabilistic selection and quick sort	1
5	Module – 5 (Approximation algorithms)	
5.1	✓ Introduction	1
5.2	✓ Vertex-cover problem	1
5.3	Traveling-salesman problem	1
5.4	Set-covering problem	1
5.5	Subset-sum problem (1)	1
5.6	Subset-sum problem (2)	1

Reference Books

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, “Introduction to Algorithms”, MIT Press, 3rd edition, 2009.
2. Gilles Brassard and Paul Bratley, “Fundamentals of algorithms”, Prentice-hall of India Private Limited, 2001.
3. Rajeev Motwani, Prabhakar Raghavan, “Randomized Algorithms”, Cambridge University Press, 2000.
4. Dexter C. Kozen, “The Design and Analysis of Algorithms”, Springer.
5. Jon Kleinberg and Eva Tardos, “Algorithm Design”, Pearson Education, 2006.